# Chapter 1 Bundles

## 1.1 Bundles

What is bundle?

### 1.1.1 Definitions

**Definition 1.1.1.1**

- In most general case, a **bundle** over an object $B$ in a category $C$ is simply an object $E$ of $C$ equipped wiht a morphism in $C$ from $E$ to $B$

$$p : E \to B$$

And let $(x : U \to B)$ be a generalized element of $B$. (**??**), the **fibre** $E_x$ of the bundle over $x$ is the pullback is the pullback $x^*E$

$$\begin{array}{ccc} x^*E = U \times_B E & \longrightarrow & E \\ \downarrow & & \downarrow p \\ U & \xrightarrow{x} & B \end{array}$$

Such bundles over $B$ is a category. Called **slide category** $C/B$ of $C$ over $B$.(We will recall this one more time in category theory)

- The morphism $E_1 \to E_2$ is a diagram such that the following diagram commutes

$$\begin{array}{ccc} E_1 & \longrightarrow & E_2 \\ & \searrow \quad \swarrow & \\ & B & \end{array}$$

of slide category.

Some of Bundles are listed as following:

- Fibre Bundle;
- Micro Bundle;
- Principle Bundle;
- Vector Bundle;
- Bundle with Connection;

**Cross Sections**

**Definition 1.1.1.2**

- A **cross section** of a bundle $(E, p, B)$ is a map $s : B \to E$, such that

$$ps = 1_B$$

In other words, a cross section is a map $s : B \to E$ such that, $s(b) \in p^{-1}(b), b \in B$.

- Furthermore, a cross subsection $(E', p', B) \subseteq (E, p, B)$, is a section such that $s(b) \in E', b \in B$.

**Product Bundles**

**Definition 1.1.1.3** The product bunlde $(B \times F, p, B)$ is a bijection

♠ **Proposition 1.1.1.4** Every cross section $s$ of a product bundle $(B \times F, p, B)$ has the form

$$s(b) = (b, f(b)) \qquad p : B \times F \to B \qquad f : B \to F$$

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∘ **Proof** Notice: $ps = 1_B \Leftrightarrow ps(b) = b \Leftrightarrow p(b, f(b)) = b$ □

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**Morphisms of Bundles**

**Definition 1.1.1.5** Let $(E, p, B)$ and $(E', p', B')$ be twon bundles, then a morphism is given by

$$(u, f) : (E, p, B) \to (E', p', B')$$

such that $p'u = fp$.

$$\begin{array}{ccc} E & \xrightarrow{u} & E' \\ \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle p'} \\ B & \xrightarrow{f} & B' \end{array}$$

And a special case is the morphism in slice category.

**Definition 1.1.1.6** Take the notion of the category of bundles

$$Bun$$

**Remark** And, the slice category $\mathcal{C}/\mathcal{B} = Bun(X)$

**Product and Sum of Bundles**

**Definition 1.1.1.7**

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**Restriction and Induced Bundles**

**Definition 1.1.1.8** Let $\xi = (E, p, B)$ be a bundle, and let $A$ be a subset of $B$.
Then, the restriction of $\xi$ to $A$, denoted by $\xi|A$, is the bundle $(E', p', A)$; $E' = p^{-1}(A), p' = p|E$.

The restriction is a map $Bun_B \to Bun_A, A \subseteq B$.

**Definition 1.1.1.9** ...

♠ **Proposition 1.1.1.10** ...

∘ **Proof** ... □

### 1.1.2 Local Properties

**Definition 1.1.2.1**

- Two bundles $\xi$ and $\eta$ over $B$ are locally isomorphic provided for each $b \in B$, there exists an open neighborhood $U$ of $b$ such that

$$\xi|U \cong \eta|U$$

- A bundle $\xi$ over $B$ is locally trivial with fibre $F$, provided $\xi$ is locally isomorphic with the product bundle $(B \times F, p, B)$.

♠ **Proposition 1.1.2.2** The relation of being locally isomorphic is an equivalence relation on the class of all bundles over $B$.

∘ **Proof** ... □

★ **Corollary 1.1.2.3** If $\xi$ is locally isomorphic to a locally trivial bundle, $\xi$ is locally trivial.

∘ **Proof** ... □

♠ **Proposition 1.1.2.4** Let $\xi$ andd $\eta$ be two bundles over $B$, with a map $f : C \to B$.
Then, if $\xi$ and $\eta$ are locally isomorphic, then $f^*(\xi)$ and $f^*(\eta)$ are locally isomorphic over $C$.

∘ **Proof** ... □

## 1.2 Homotopies,and CW-Complexes

▶ ***Theorem*** $\xi = (E, p, B)$ be a locally trivial bundle, with fibre $F$, where $(B, A)$ is a relative CW-complex. Then, all cross sections $s$ of $\xi|A$ prolong to a cross section $s^*$ of $\xi$, under either of following hypothesis:

- The space $F$ is $(m-1)$-connected for each $m \leq dim(B)$;
- There is a relative CW-complex $(Y, X)$ such that

$$B = Y \times I \qquad A = (X \times I) \cap (Y \times 0) \qquad I = [0, 1]$$

∘ **Proof** ... □

## 1.3 Connections

# Chapter 2 Fibre Bundles

## 2.1 Fibre Bundles

**Definition 2.1.0.1** Let $\eta : E \to B$ be a bundle. Then, a **fiber bundle** is

$$(\eta, E, B, F)$$

with a topological space called: standard fiber $F$, such that for every general element $1 \to B$, the pullback of $E$ along $x$ is isomorphic to $F$.
i.e. $E_x = \eta^{-1}(x) \cong F$.

Equivalently, there exists factorization

$$F \to E \overset{\eta}{\to} X$$

Such $F \to E$ is not canonical: Choose a base point $x_0 \in X$, thus there exists factorization

$$F \overset{\cong}{\to} E_{x_0} \hookrightarrow E$$

# Chapter 3 Associated Bundles

# Chapter 4 Principal $G$-Bundles and Classifying Space

## 4.1 Principal $G$-Bundles

**Definition 4.1.0.1** For $G$ a group, and $X$ some other object.
A $G$-principal bundle over $X$, also called $G$-torsor over $X$, is a bundle $P \to X$ equipped with $G$-action

$$\rho : P \times G \to P$$

such that

- The action is pricipal: The shear map

$$(p_1, \rho) : P \times G \to P \times_X P \qquad (s,t) \mapsto (p_1(s), \rho(t))$$

is an isomorphism. And the free action and transitive

$$\rho(s,t) = s \Rightarrow t = e \qquad \rho(s, t_1 t_2) = \rho(\rho(s,t_1), t_2)$$

- or equivalently,

$$(P \to X) \cong (P \to P/G)$$

- or, with the cotinuous function $p : P \to X$ and the action $\rho : P \times G \to P$ on $G$ over $X$, fitting into coequalizing diagram

$$\begin{array}{c} P \times G \\ {\scriptstyle\rho}\downarrow\downarrow{\scriptstyle p_1} \\ P \\ \downarrow{\scriptstyle p} \\ X \end{array}$$

and this is locally trivial in the sense that

$$\begin{array}{ccccc} U \times G & \xleftarrow[\simeq_G]{p_1} & P|U & \longrightarrow & P \\ & \searrow & \downarrow{\scriptstyle p|U} & & \downarrow{\scriptstyle p} \\ & & U & \longrightarrow & X \end{array}$$

We will firstly show

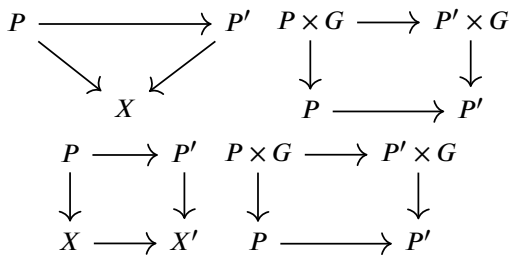
♠ **Proposition 4.1.0.2** The definitions are equivalent.

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∘ **Proof** ... □

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**Definition 4.1.0.3** The morphisms between principal $G$-bundles is specified by



**Pullback**

♠ **Proposition 4.1.0.4** ...
◇ **Lemma 4.1.0.5** For every $G$-bundle morphism $(F, f)$, there exists a natural isomorphism

$$P \cong f^*Q$$

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∘ **Proof** By universal property, there exists a mapping

$$\Phi : P \to f^*Q$$

Then:

- It is well-defined:
- It is $G$-equivariant:
- $\Phi$ is a homeomorphism:

Now, via local trivialization, on $U \subseteq X$,

$$P|U \cong U \times G \qquad (f^*Q)|U \cong U \times G$$

□

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### 4.1.1 Groupoind over $G$-Bundles

♠ **Proposition 4.1.1.1** TFAE: For

$$\begin{array}{ccc} P & \xrightarrow{F} & Q \\ \downarrow{\scriptstyle p} & & \downarrow{\scriptstyle q} \\ X & \xrightarrow{f} & Y \end{array}$$

- $(F, f)$ is inversable.
- $f$ is a homeomorphism

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∘ **Proof** ⇒ Proved;
⇐ If $f : X \to Y$ is a homeomorphism, the projection

$$\pi_2 : f^*Q \to Q$$

is also a homeomorphism:
And, $F = \pi_2 \circ \Phi$ is also a homeomorphism. Thus,

$$(F, f)^{-1} = (F^{-1}, f^{-1})$$

□

♠ **Proposition 4.1.1.2** $Bun_G(X)$ is actually a groupoid.

∘ **Proof** By above. □

**Remark** $Bund_G$ may be not groupoid!!

### 4.1.2 Degree-1 Cohomology

♠ **Proposition 4.1.2.1** Let $G$ be a topological space. Then, there exists isomorphism

$$H^1(X, \underline{G}) \simeq GBund(X)/\simeq = \pi_0 \operatorname{Bun}_G(X) \qquad \underline{G}(U) = C(U, G)$$

between the degree-1 $G$ cohomology of $X$, and the isomorphism classes of $G$-principal bundles over $X$.
**Remark** Here, the cohomology is the Cech cohomology.

∘ **Proof** ... □

Furthermore:

♠ **Proposition 4.1.2.2** If $X$ can CW-approxiamated, then

$$H^1(X, \underline{G}) \cong H^2(X; \mathbb{Z})$$

∘ **Proof** ... □

## 4.2 Classyfing Space

### 4.2.1 Grassmannians, and Infinite Grassmannians, Stiefel Manifold

**Definition 4.2.1.1** ...

### 4.2.2 title

**Definition 4.2.2.1** For $G$ a topological group, there is a classifying space $BG \in Top$, for topological $G$-principal bundles. With the following natural isomorphsim

$$GBund(X)_0 \simeq \pi_0 Top(X, BG)$$

### 4.2.3 Milnor's Construction of Universal Bundle

▶ ***Theorem*** (Milnor) Let $G$ be a topological space. Then, there exists factorization

$$G \to EG \to BG$$

where:

- $EG$ is constractible;

- $G$ acts freely on $G$, and $p$ is the locally trivial principal bundle;

Furthermore, if $X$ is a paracompact Hausddorff space, then

$$[X, BG] \cong \pi_0 Bun_G(X)$$

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∘ **Proof** ... □

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### 4.2.4 Stiefel Construction

# Chapter 5 Projective Modules and Vector Bundles

## 5.1 Free module, $GL_n$ and Stably Free Modules

### 5.1.1 The Module over Rings

Review:

- Vector Space(over division ring);
- Base;
- IBP property;

  ◇ **Lemma 5.1.1.1** If there exists a ring map $f : R \to F$, where $F$ is a field or division ring, then $R$ has IBP property.

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∘ **Proof** Consider the following map

$$M \to M \otimes_R F = V \qquad m \mapsto m \otimes_R 1$$

Then, $\sum \lambda_i m_i = 0 \to \lambda_i = 0 \Leftrightarrow \sum \lambda_i (m_i \otimes_R 1) = 0 \to \lambda_i = 0$.
Especially, if commutative, we could take $F = R/m$. □

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Thus, $dim(V) = dim(M)$ is well-defined.

**Definition 5.1.1.2** In general, for the ring $R$, take the notation of

$$GL_n(R)$$

for the group of invertible matrices. Especially, $n = 1$, $GL_1(R) = R^\times$.

◇ **Lemma 5.1.1.3** Any finite-dimensional algebra over a field/division ring must satisfies $IBP$, because the rank of free $R$-module $M$ is an invariant:

$$rank(M) = dim_F(M)/dim_F(R)$$

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∘ **Proof** ... □

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### 5.1.2 Stably Free

**Definition 5.1.2.1** An $R$-module $P$ is called **stably free** of rank $n-m$ if

$$P \oplus R^m \cong R^n$$

for some $m, n$.
Conversely:

♣ **Example 5.1.2.2** ...

### 5.1.3 Stably Free Property

**Definition 5.1.3.1** ...

### 5.1.4 Some of Noncommutative Ring Theory

♠ **Proposition 5.1.4.1** (P.M.Cohn) Consider the following conditions on a ring $R$:

- $R$ satisfies IBP;
- For all $m, n$, if $R^m \cong R^n \oplus P$, then $m \geq n$;
- For all $n$, if $R^n \cong R^n \oplus P$, then $P = 0$.

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∘ **Proof** ... □

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## 5.2 Projective Modules

Given in noncommutative ring theory/Homological Algebra

## 5.3 Picard Group of Commutative Rings

Given in noncommutative ring theory.

## 5.4 Topological Vector Bundle, and Chern Classes

We will work on field $F = \mathbb{R}, \mathbb{C}$ or $\mathbb{H}$.

### 5.4.1 Families of Vector Spaces

**Definition 5.4.1.1**

- Let $X$ be a topological space. A family of vector spaces is a topological space $E$, together with map

  $$\eta : E \to X$$

  with a finite dimensional vector space structure on each $E_x = \eta^{-1}(\{x\})$.
  The structure of $E_x$ is compatible with the topological structure of $E$. i.e.

  $$F \times E \to E \qquad E \times_X E \to E$$

  are continuous.

- A **homomorphism** from $\eta : E \to X$ to $\varphi : F \to X$ is:
  - A continuous map $f : E \to F, \eta = \varphi f$;
  - $f_x : E_x \to F_x$ is a linear map of vector spaces;

If $Y \subseteq X$, then define

$$E|Y = (E|Y, Y, \eta|\eta^{-1}(Y))$$

With restriction map $\eta|\eta^{-1}(Y) : \eta^{-1}(Y) \to Y$.

In general:

**Definition 5.4.1.2** If $f : Y \to X$ is any continuous map, then the **pullback** of $\eta : E \to X$ is

$$f^*\eta : f^*E \to Y$$

where $f^*E = Y \times_X E$

$$\begin{array}{ccc} f^*E & \longrightarrow & E \\ \downarrow{\scriptstyle f^*\eta} & & \downarrow{\scriptstyle \eta} \\ Y & \xrightarrow{f} & X \end{array}$$

**Vector Bundles**

♣ **Example 5.4.1.3** (Trivial Bundle) If $V$ is an $n$-dimensional vector space. The projection from $T^n = X \times V$ to $X$ is a **constant family**

$$\eta : X \times V \to X$$

Such bundle is called **trivial vector bundle**.

**Definition 5.4.1.4** (Vector Bundle) A **vector bundle** is a family of vector spaces such that:

- Local Trivial: For every point $x \in X$, $\eta|U$ is a trivial.

$$\begin{array}{ccc} \eta^{-1}(U) & \rightleftarrows & T = U \times V \\ & \searrow \quad \swarrow & \\ & U & \end{array}$$

So sometimes a vector bundle is called a **locally trivial** family of vectors.

**Definition 5.4.1.5** (Category of vector bundles) Define the category $Vect_F(X)$, with the elements:

- $Ob(Vect_F(X)) = \{\text{Vector Bundles on } X\}$
- $Mor(Vect_F(X)) = \{\text{Morphisms between elements in } Ob(Vect_F(X))\}$

And a functor is given by: $f^* : Vect(Y) \to Vect(X)$.

**Product**

**Definition 5.4.1.6** Let $\eta : E \to X, \varphi : F \to X$ be two vector bundle. Then, the **product of vector bundles** is defined as $(E \times F, X, \eta \times \varphi)$

$$\eta \times \varphi : E \times F \to X \times X$$

This gives a bilinear functor

$$\boxplus : Vect(X) \times Vect(X) \to Vect(X \times X)$$

**Tensor Product**

**Definition 5.4.1.7** Let $\eta : E \to X, \varphi : F \to X$ be two vector bundles. Then,

$$E \otimes F := \coprod_{x \in X} E_x \otimes F_x$$

with $\eta \otimes \varphi : E \otimes F \to X, e \otimes f \mapsto x; e \in E_x, f \in F_x$.

This is also a bilinear functor

$$\otimes : Vect(X) \times Vect(X) \to Vect(X)$$

**Whitney Sum**

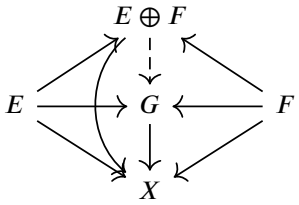
**Definition 5.4.1.8** The **Whitney sum** $E \oplus F$ of two vector bundle $\eta : E \to X, \varphi : F \to X$ is the family of all vector spaces $E_x \oplus F_x$, which can be seen as a topological subspace of $E \times F$:

$$(E_x \times F_x)_{x \in X} \hookrightarrow E \times F$$

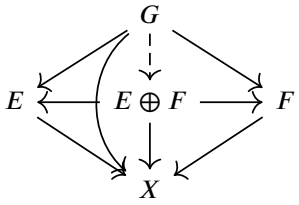
With the new vector bundle construction induced by $\eta \times \varphi$:

$$\eta \oplus \varphi := \eta \times \varphi | E \oplus F$$

Whitney sum is the categoricial coporudct in $Vect(X)$.




Given $E \oplus F \to G : E_x \times F_x \ni (e, f) \mapsto i_1(e) + i_2(f)$ which is well-defined.
And commutes: Via the product structure.
Especially: $E \oplus F$ is also the categoricial coproduct:




$G \to E \oplus F, g \mapsto p_1(g) + p_2(g)$. Omitted.

★ **Corollary 5.4.1.9** $Vect(X)$ is an additional category.

∘ **Proof** Via checking definition. □

♣ **Example 5.4.1.10** (Mobius Bundles) The Mobius bundle $E^1$ over $S$

$$\pi : E^1 \to S$$

With strip $E_x \cong \mathbb{R}, x \in S^1$.

$E^1$ can be seen as the pasting of $[0,1] \times \mathbb{R}$ via

$$(0, v) \sim (1, -v)$$

♣ **Example 5.4.1.11** (Whitney Sum of Tangent Bundle of Product Manifolds) For the product of manifolds $M_1 \times M_2$, we have

$$T_{(x,y)}(M \times N) \cong T_x M \oplus T_y N$$

Thus, we have

$$T(M \times N) = TM \oplus TN \to M \times N$$

New Vector Bundles from Olds:

- Restsriction;
- Induced;
- Carstian Product
- Whitney Sum;

**Diemension of Vector Bundles**

Notice that: $dim(E) : X \to \mathbb{N}, x \mapsto E_x$ is locally constant.

**Definition 5.4.1.12** An $n$-dimensional vector space is a bundle $E$ such that $dim(E) = n$ is a constant function. Especially, on connected $X$, the function is constant.

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**Definition 5.4.1.13** ...

**Definition 5.4.1.14** Define:

$$V_{n,F}(X) :=$$

**Subbundles**

**Definition 5.4.1.15**

- A **subbundle** of a vector space $\eta : E \to X$ is a subspace $F$ of $E$ which is a vector bundle under induced structure.

  i.e. $F_x$ is a vector subspace of $E_x$, and family $F \to X$ is locally trivial.

- The **quotient bundle** $E/F$ is defined as the union of vector spaces

$$(E_x/F_x)$$

given quotient topology(by $\eta$).

Short exact sequence

$$0 \longrightarrow F \longrightarrow E \longrightarrow E/F \longrightarrow 0$$

in $VB(X)$.

**Definition 5.4.1.16** A **vector bundle** $E \to X$ is said to be of **finite type** is there is a finite covering $U_1, \dots, U_n$ such that $E|U_i$ is a trivial bundle.

♠ **Proposition 5.4.1.17**

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∘ **Proof** ... □

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▶ ***Theorem*** (Subbundle Theorem) Let $E \to X$ be a vector bundle on paracomapct space $X$:

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∘ **Proof** With the conclusion 5.4.1, there exists a metric on the bundle. □

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**Sections**

**Definition 5.4.1.18**

- Let $\eta : E \to X$ be a vector space. A **global section** of $\eta$ is a continuous map $s : X \to E$ such that

$$\eta \circ s = 1_X$$

- It is nowhere zero if $s(x) \neq 0, \forall x \in X$.

In general, we could define the **local section**

$$s : U \to E \qquad \eta \circ s = id_U$$

**Vector Bundle under Riemannian Metrics**

Review:

- Riemannian Metrics;
- Hermitian Metrics;

Metric on $E_x \Rightarrow$ Metric on $E$.
A fundamental result is:

▶ ***Theorem*** If $X$ is a paracompact space, then there exists a metric on $E$, when $F = \mathbb{R}, \mathbb{C}, \mathbb{H}$.

◦ **Proof** ... □

**Patching Vector Bundles**

Transition of vector bundles: Consider two locally trivial maps

...

**Definition 5.4.1.19** (Tensor Product of Bundles)

**Definition 5.4.1.20** (Determinant Bundle)

**Cancellation Theorems**

▶ ***Theorem*** (Real Cancellation Theorm)

◦ **Proof** ... □

**Homotopy Invariance Theory**

**Classifying Vector Bundles**

▶ ***Theorem*** (Classifying Theorem) Let $X$ be paracompact space. Then,

$$VB_n(X) \cong [X, Grass_n]$$

◦ **Proof** ... □

**Definition 5.4.1.21** (Classifying Spaces)

▶ ***Theorem*** (Classifying Theorem for Line Bundles)

...

◦ **Proof** ... □

### 5.4.2 Serre-Swan Theorem

♠ **Proposition 5.4.2.1** (Global Section) Let $\eta : E \to X$ be a vector bundle. Let $\Gamma(E)$ be the set of global sections of $E$.
Then, $\Gamma(E)$ is a module over $C^0(X)$ of continuous functions on $X$.
Especially, if $E$ is a trivial bundle of dimension $n$, then $\Gamma(E)$ is a free $C^0(X)$-module of rank $n$.

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∘ **Proof** Module axiom:

- Sum:
- Scalar Product:

□

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★ **Corollary 5.4.2.2** If $X$ is paracompact, then $\Gamma(E)$ is a locally free $C^0(X)$-module.
$\Gamma(E)$ is a finitely generated projective module if $X$ is compact or $E$ is of finite type.

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∘ **Proof** ... □

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▶ ***Theorem*** (Swan) Let $X$ be a compact Hausdorff space, and write $R$ for $C^0(X)$.
Then, $\Gamma : VB(X) \to P(R)$ of finitely generated projective modules is a functor. The homomorphism

$$\Gamma_{VB(X)}(E, F)$$

This implies, there exists an equivalence of categories $VB(X) \cong P(C^0(X))$.

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∘ **Proof** ... □

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**Remark** The Serre's theorem will be given in AG.
Here:

- Construction:
- Affines:

**Projective Flag Modules**

**Splitting Principle**

**Pontrijagin Classes**

### 5.4.3 Characteristic Classes

**Abstract** The determinant line bundle $det(E)$ of a vector bundle $E$ yields to a cohomology class.

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**Axioms for Stiefel-Whitney Classes**

† **Axiom** ...

**Axioms for Chern Classes**

† **Axiom** ...

**Pontrjagin Classes**

**Definition 5.4.3.1** (Complex Conjugate Bundle)

### 5.4.4 Others

**Complexification of Real Vector Bundlees**

## 5.5 Algebraic Vector Bundles